

MEHMOOD UL HASSAN

Senior Backend Engineer · Distributed Systems, Python & Cloud-Native Infrastructure

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PROFILE

Backend engineer with 5+ years building distributed systems in Python (FastAPI, asyncio) and Node.js/NestJS, with services communicating over gRPC/Protobuf, Kafka and RabbitMQ. Led backend teams of up to six engineers, built and operated Kubernetes infrastructure on GKE, and cut server response times by ~40% through Redis caching and database optimization. CKAD certified. Working knowledge of Go; focused on performance, horizontal scaling and reliability.

CERTIFICATIONS

- **Certified Kubernetes Application Developer (CKAD)** · The Linux Foundation / CNCF · Sep 2026 · ID LF-sxz34xuhog · [Credly badge](#) · [LF verification](#)
- **Salesforce Certified B2C Commerce Developer** · [View](#)

Courses

- SQL for Data Science · Coursera (UC Davis) · [View](#)
- Introduction to Structured Query Language (SQL) · Coursera (University of Michigan) · [View](#)
- HTML, CSS & JavaScript for Web Developers · Coursera (Johns Hopkins University) · [View](#)

PROFESSIONAL EXPERIENCE

Lead Full Stack Cloud Engineer · Code Encoders

Dec 2024 – Present

Lahore, Pakistan

- Lead backend architecture for a microservices platform built on FastAPI and Python async patterns; own service design, PR review and engineering standards, and mentor the development team.
- Built GKE Autopilot infrastructure from scratch on a private VPC, designed for horizontally scaled, multi-environment deployments.
- Deployed services communicating over gRPC and TCP, each with its own database in a private subnet; added targeted indexing to cut query times.
- Implemented Redis caching to reduce database load and led a refactor that cut server response times by ~40%.
- Rolled out self-hosted API gateways with DNS-based ingress for high availability and traffic routing.
- Configured Google Cloud Armor rate limiting and security policies to protect services against DDoS attacks.
- Built CI/CD pipelines with GitHub Actions and Jenkins; lead production troubleshooting to keep services available.

Stack: Python, FastAPI, NestJS, Node.js, gRPC, RabbitMQ, Redis, MongoDB, GCP (GKE Autopilot, VPC, Cloud Armor), Kubernetes, GitHub Actions, Jenkins, IaC

Freelance Backend Engineer (Contract) · Upwork

2026 · 2 months

Remote

- Delivered an end-to-end Bill of Lading platform for a logistics client in 8 weeks, from invoice intake to a bank-ready document pack.
- Integrated the Kimi LLM to extract shipper, buyer, goods and weight data from commercial invoices, with operator review and correction in a gated 5-step wizard.
- Built a rendering pipeline that fills each shipping line's Excel BL template and converts it to PDF with headless LibreOffice; previews are always regenerated from live data and watermarked DRAFT.
- Designed a tamper-evident approval flow: approved artifacts are written once, checksummed and versioned (new approvals supersede, never overwrite), and files failing checksum verification are refused at serve time.
- Added a verification link on approved documents so banks can confirm authenticity, and generated a zipped bank submission pack with a manifest.

Lead Backend Engineer • Consforc LLC

Jul 2023 – Sep 2024

Boston / New York, USA (Remote)

- Led backend development of a Python/FastAPI microservices platform and managed a 6-member backend team, including PR review and delivery.
- Designed the platform around per-service databases to isolate load and allow services to scale independently.
- Implemented gRPC for low-latency internal communication and Kafka for high-volume asynchronous messaging.
- Built a custom GraphQL gateway in NestJS that routed client requests across services.
- Integrated Redis caching, containerized services with Docker and maintained Kubernetes clusters for deployment and scaling.
- Managed CI/CD and project delivery in Azure DevOps, improving release consistency.

Stack: Python, FastAPI, NestJS, GraphQL, gRPC, Kafka, Redis, Docker, Kubernetes, Azure

Lead Backend Engineer • Otobucks

Jan 2023 – Sep 2023

UAE (Remote)

- Led backend development with Python, FastAPI, Redis and PostgreSQL.
- Designed MongoDB data models using aggregation pipelines and virtual relations.
- Built and optimized the Service Provider Panel, Admin Panel and Android app backend; guided developers and defined code-quality standards.

Stack: Python, FastAPI, PostgreSQL, Redis, MongoDB, MySQL, Node.js, Express, AWS

Lead Software Engineer • Eguana Commerce

Dec 2022 – Aug 2023

Seoul, South Korea (Remote)

- Led Under Armour's transition to headless commerce on Salesforce Commerce Cloud, redesigning the product listing page for scalability and performance.
- Migrated cartridges from SiteGenesis to SFRA; built Impex jobs, services and the TossPay payment integration.
- Implemented Return Merchandise Authorization and Personal Information Protection compliance (consents, action logs); reviewed PRs and managed deployments.

Stack: SFCC, SFRA, Cartridges, Jobs, Services, JS/ISML

Software Engineer • Aiva Creative

Jun 2021 – Dec 2022

Lahore, Pakistan (Hybrid)

- Built SFCC backend features across multiple B2B/B2C brands, including OCAPI integrations.
- Delivered L'Oréal Japan integrations for Braintree, PayPal, Yotpo and GTM, plus Impex jobs.
- Worked across PDP, PLP, Cart and Checkout with a focus on SEO and server-side rendering performance.

Stack: SFCC, Custom Cartridges, JavaScript, ISML/XML, AJAX

TECHNICAL SKILLS

Languages	Python, Go, JavaScript, TypeScript
Backend	FastAPI, asyncio, Django, NestJS, Node.js, Express, Fastify
Data & Caching	PostgreSQL, MySQL, MongoDB, Redis
APIs & Messaging	gRPC / Protobuf, REST, GraphQL, Kafka, RabbitMQ, TCP
Cloud & Infra	GCP (GKE Autopilot, VPC, Cloud Armor), AWS (Lambda, EKS), Azure, Kubernetes, Helm, Docker, IaC
CI/CD	GitHub Actions, Jenkins, Azure DevOps
Other	System design, microservices, AI integrations (Kimi, Hume AI, AssemblyAI), Salesforce Commerce Cloud

EDUCATION

BS Software Engineering • Virtual University of Pakistan, Lahore

2016 – 2020

INTERESTS

Competitive football: attacking midfielder/forward for Real Lahore FC; formerly Atlético de Madrid Academy Pakistan (U-19).